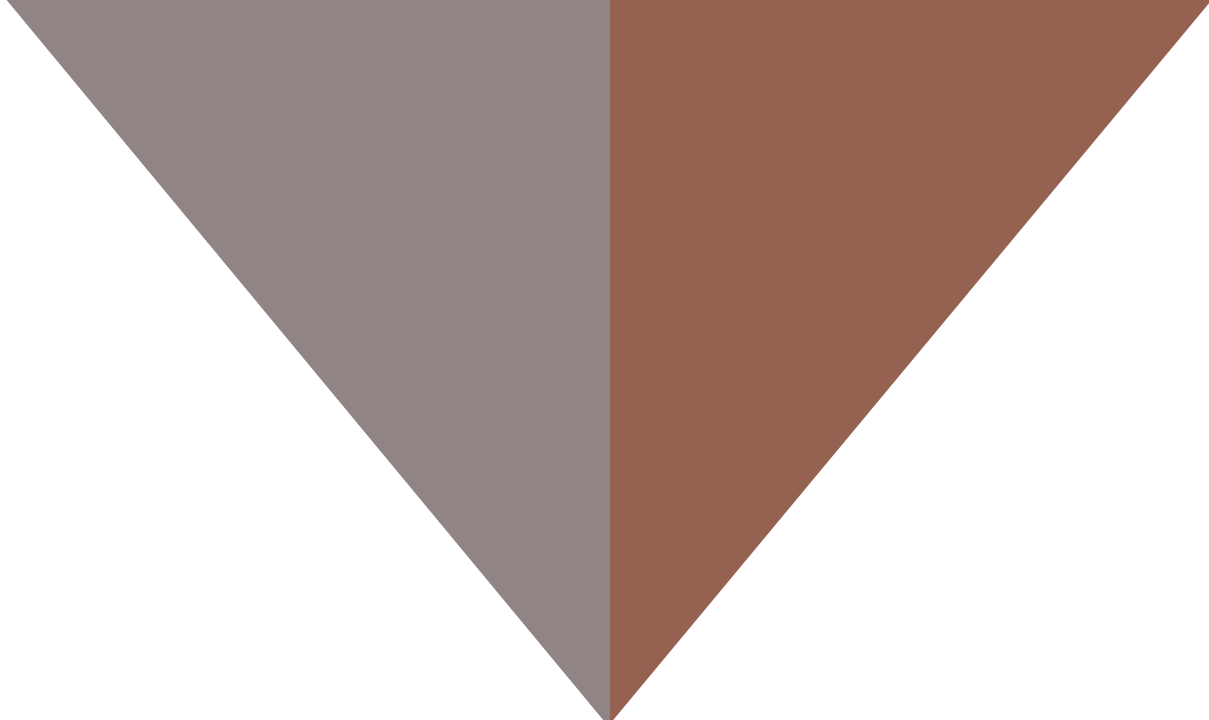


ROBUSTNESS & RESILIENCY OF MULTI-AGENT SYSTEMS UNDER ATTACKS AND DISTURBANCES

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AGENDA

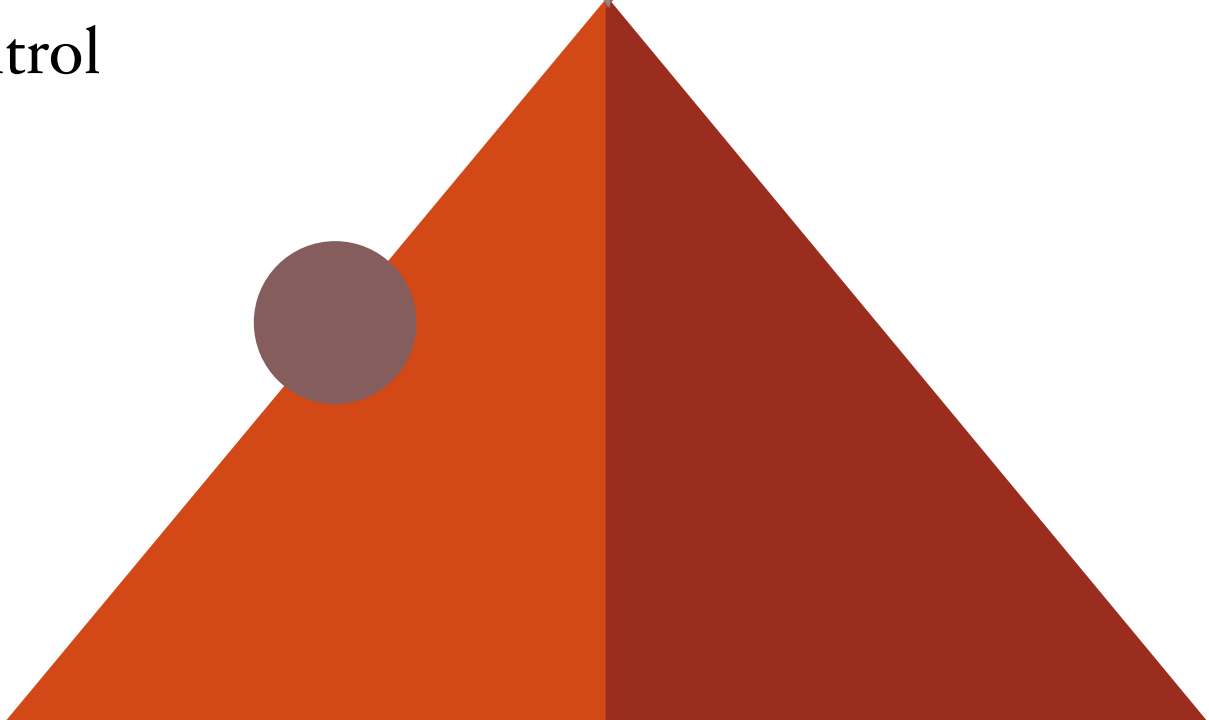
Introduction

Robust and Resilient Distributed Control

Simulation

Results

Summary



INTRODUCTION

- Multi-Agent systems can be presented by graphs consisting of nodes (agents) and links (communications).
- Cyber-attacks and Disturbances on an agent could affect all other agents; and system's overall performance.
- Combination of H_∞ and Adaptive controller mitigates the effects of these attacks and disturbances so we could achieve synchronization to leader's states.

LEADER-FOLLOWER SYNCHRONIZATION

- N agents, with dynamics given as:

$$\dot{\mathbf{x}}_i(t) = \mathbf{A}\mathbf{x}_i(t) + \mathbf{B}\mathbf{u}_i(t) + \mathbf{D}\mathbf{w}_i(t)$$

- Leader's states satisfy:

$$\dot{\mathbf{x}}_0(t) = \mathbf{A}\mathbf{x}_0(t)$$

- Local neighborhood error of node i as:

$$\delta_i = \sum_{j \in \mathcal{N}_i} e_{ij}(x_j - x_i) + g_i(x_0 - x_i)$$

LEADER-FOLLOWER SYNCHRONIZATION

- Distributed Control Protocol:

$$\mathbf{u}_i = cK\delta_i$$

- Synchronization error:

$$\varepsilon = \mathbf{x} - \underline{\mathbf{x}}_0$$

- Bounded L_2 -Gain Synchronization Problem:

$$\int_0^\infty \varepsilon^T Q \varepsilon \, dt \leq \gamma^2 \int_0^\infty \mathbf{w}^T W \mathbf{w} \, dt$$

ATTACK TYPE AND MODELING

- Attack on actuators could be modeled as:

$$u_i^c = u_i + \alpha_i u_i^a$$

- System under attack:

$$\dot{x}_i = Ax_i^c + Bu_i^c + Dw_i$$

PREDICTOR

$$\begin{cases} \dot{\hat{x}}_i = A\hat{x}_i + cBK \left(\sum_{j \in \mathcal{N}_i} e_{ij}(\hat{x}_j - \hat{x}_i) + g_i(x_0 - \hat{x}_i) \right) \\ \hat{x}_i(0) = x_i(0) \end{cases}$$

ROBUST AND RESILIENT CONTROL PROTOCOL

$$u_i(t) = u_{0,i}(t) + u_{1,i}(t)$$

- $u_{0,i}(t)$ is a H_∞ controller assuring Robust Performance.
- $u_{1,i}(t)$ is an adaptive controller using predictor's information to reject attacks.

$$\mathbf{u}_i(t) = \mathbf{u}_{0,i}(t) + \mathbf{u}_{1,i}(t) = cK\delta_i - d_i$$

$$\dot{d}_i = \theta cK(\hat{\delta}_i - \delta_i) - \theta d_i$$

PARAMETERS

- Coupling gain c :

$$c > \frac{1}{2\rho} > 0$$

$$\rho = \min \operatorname{Re}(\lambda_i)$$

- λ_i is the i -th eigenvalue of $L + G$ matrix.

PARAMETERS

- Feedback control gain matrix K :

$$K = R_1^{-1} B^T P_1$$

$$R_1 = R_1^T > 0$$

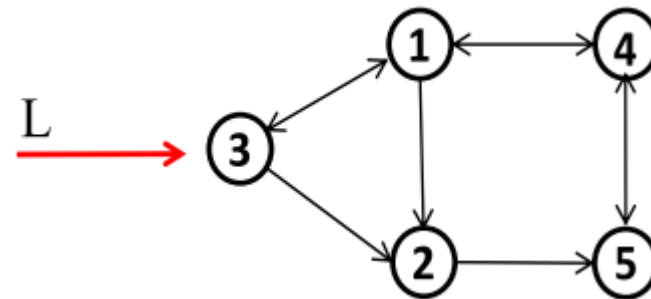
$$P_1 = P_1^T$$

$$A^T P_1 + P_1 A + Q_1 - P_1 B R_1^{-1} B^T P_1 + \frac{1}{\gamma^2} P_1 D D^T P_1 = 0$$

$$Q_1 = Q_1^T$$

SIMULATION

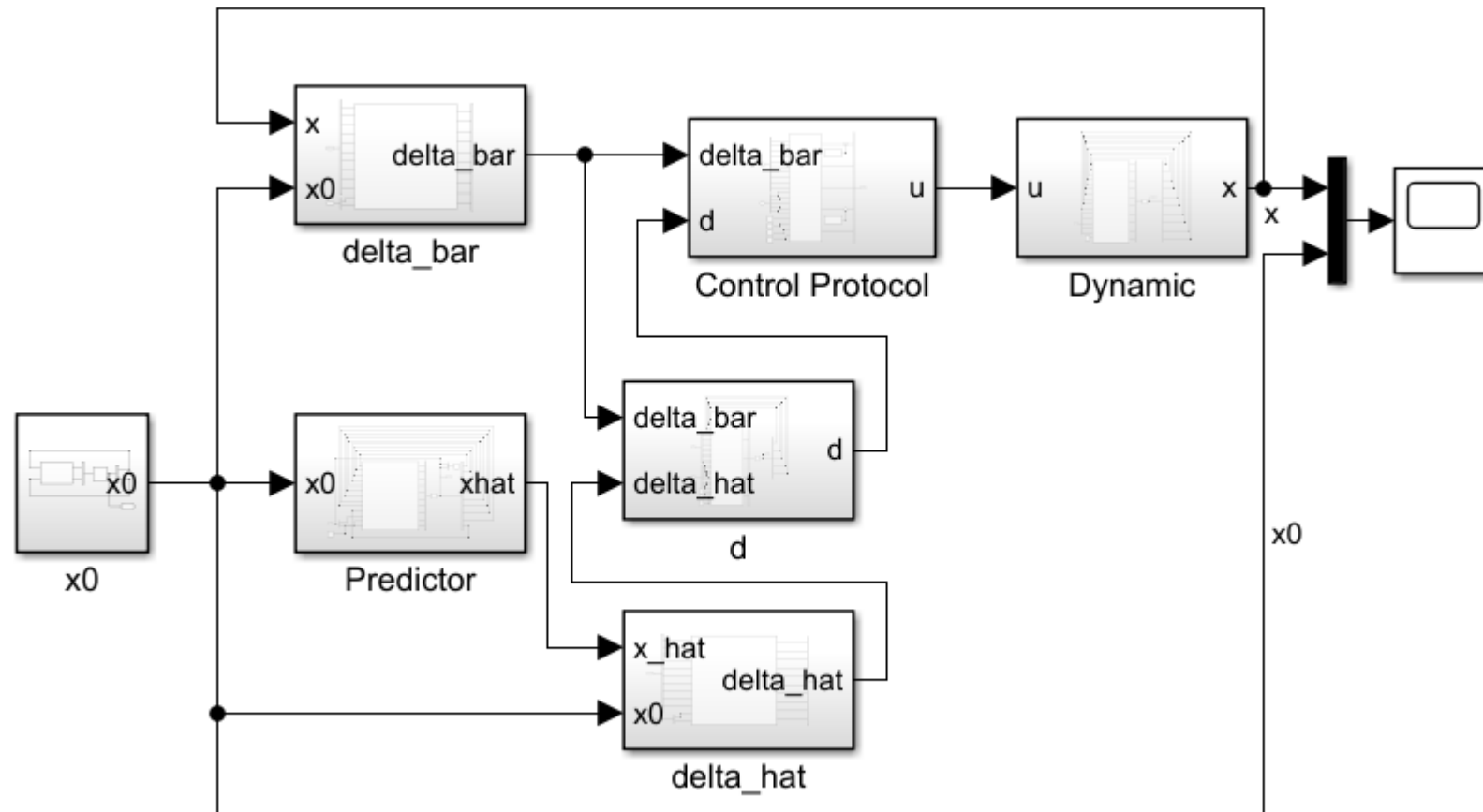
GRAPH



DYNAMIC

- $\dot{x}_i = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} x_i + \begin{bmatrix} 2 \\ 1 \end{bmatrix} u$
- $u_1^a = 1$
- $u_4^a = \sin(t)$
- $w_1 = 0.1 \sin(10t)$
- Q_1 and R_1 identity

SIMULATION

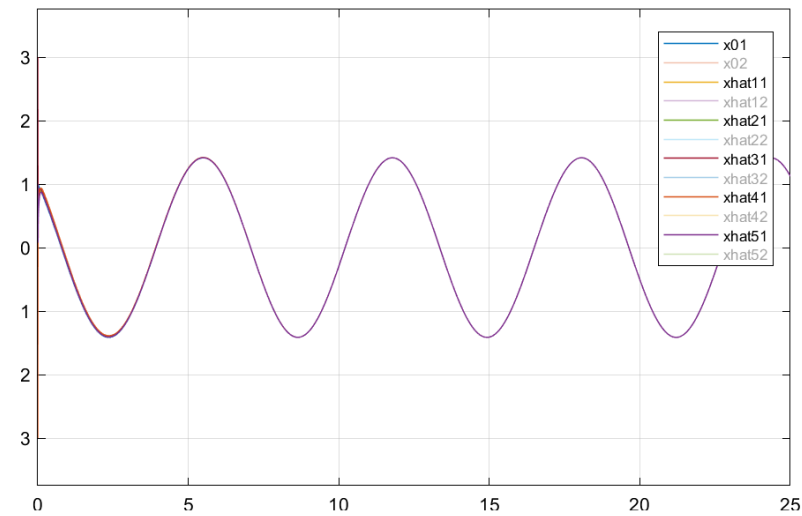


SIMULATION PARAMETERS

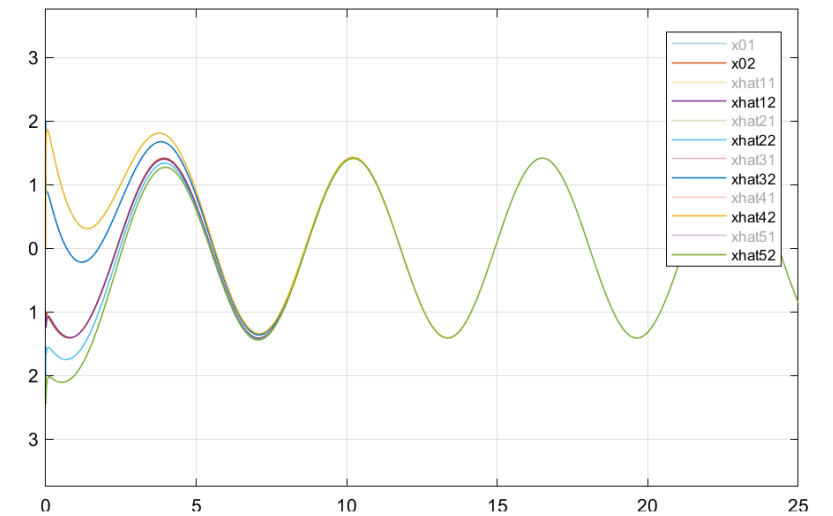
Parameter	Value
γ	60
θ	20
c	100
P_1	$\begin{bmatrix} 0.96 & -0.5 \\ -0.5 & 0.98 \end{bmatrix}$
K	$[1.42 \quad -0.02]$
$x_0(0)$	$[1; -1]$
$x(0)$	$[1; -1; 0; -2; 3; 2; -3; 0; 1; -2]$
$\hat{x}(0)$	$[1; -1; 0; -2; 3; 2; -3; 0; 1; -2]$

RESULTS

PREDICTOR'S STATE x_{i1}
 SYNCHRONIZE TO LEADER'S STATE

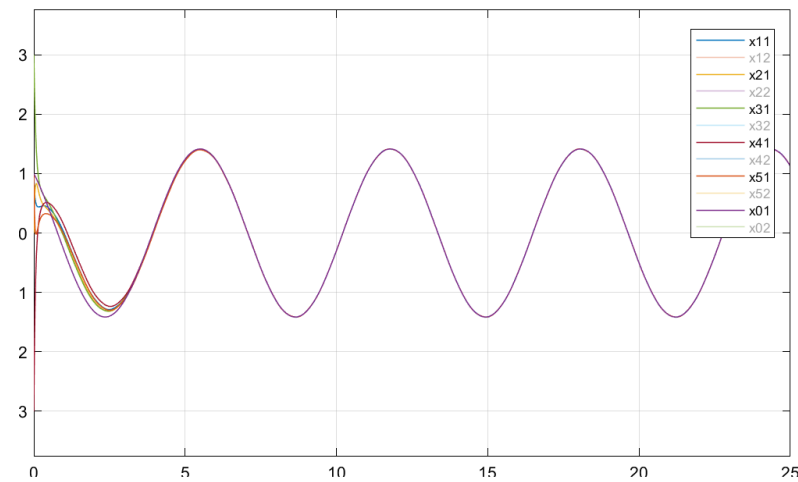


PREDICTOR'S STATE x_{i2}
 SYNCHRONIZE TO LEADER'S STATE

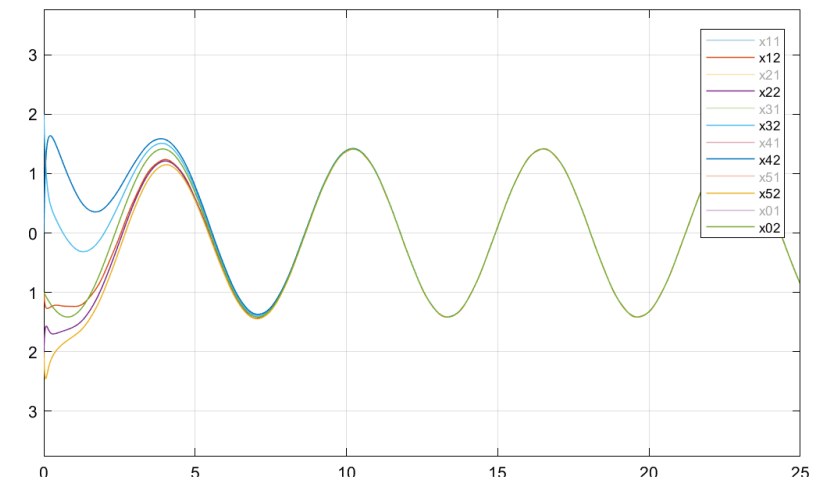


RESULTS

STATE x_{i1} SYNCHRONIZE TO
LEADER'S STATE WHEN $\alpha = 0$
AND $c = 3$

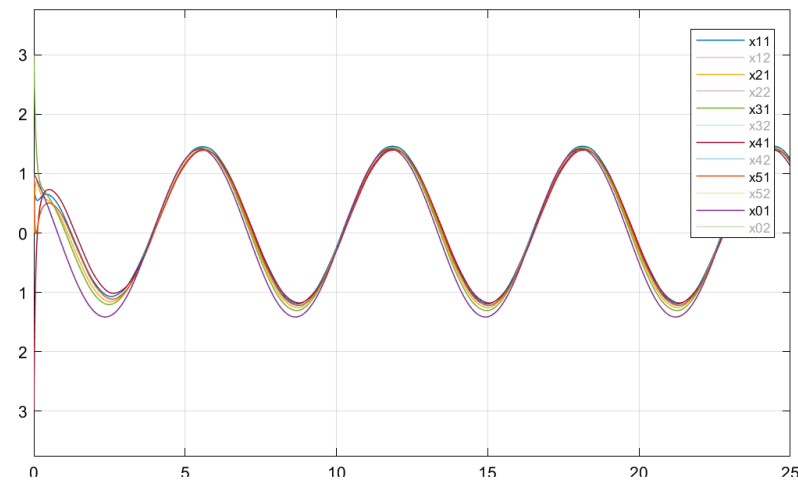


STATE x_{i2} SYNCHRONIZE TO
LEADER'S STATE WHEN $\alpha = 0$
AND $c = 3$

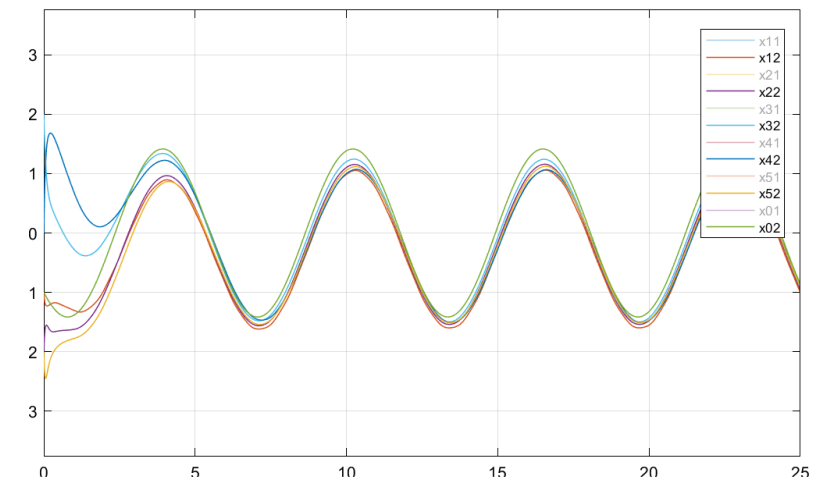


RESULTS

STATE x_{i1} SYNCHRONIZE TO
LEADER'S STATE WHEN $\alpha = 1$
AND $c = 3$

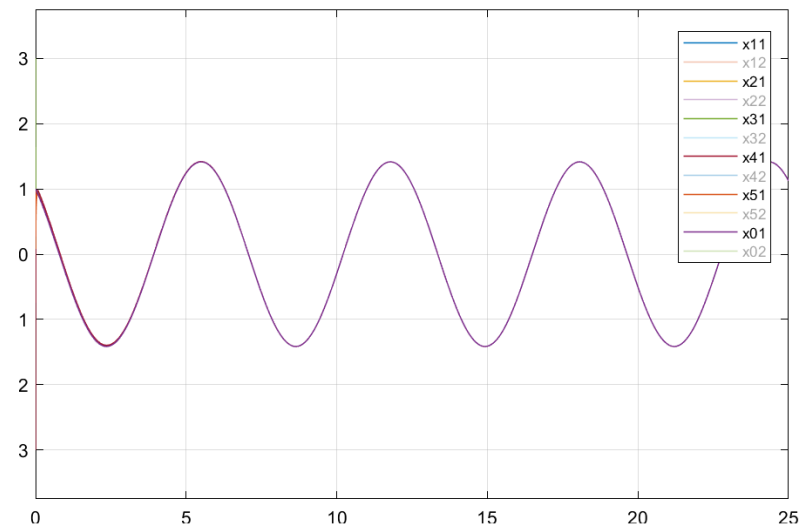


STATE x_{i2} SYNCHRONIZE TO
LEADER'S STATE WHEN $\alpha = 1$
AND $c = 3$

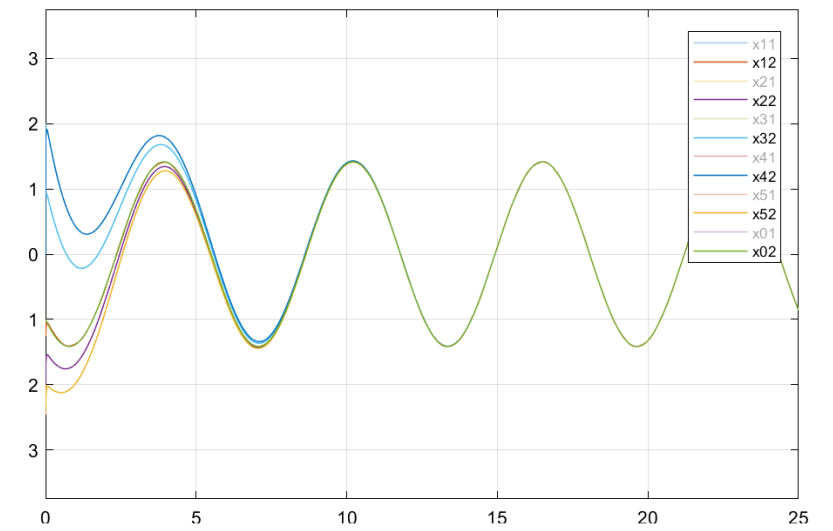


RESULTS

STATE x_{i1} SYNCHRONIZE TO
LEADER'S STATE WHEN $\alpha = 0$
AND $c = 100$

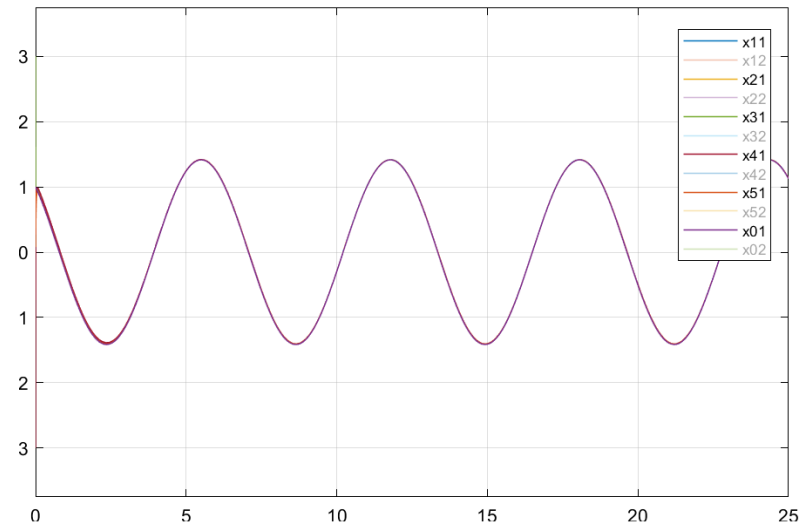


STATE x_{i2} SYNCHRONIZE TO
LEADER'S STATE WHEN $\alpha = 0$
AND $c = 100$

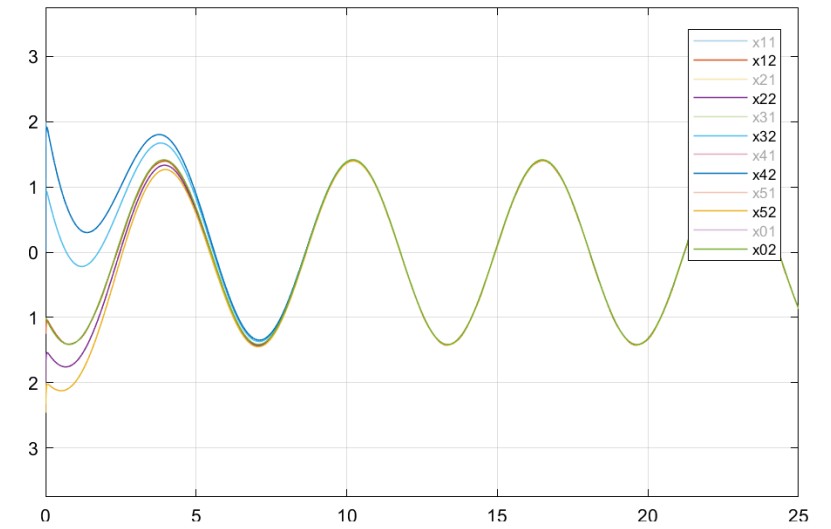


RESULTS

STATE x_{i1} SYNCHRONIZE TO
LEADER'S STATE WHEN $\alpha = 1$
AND $c = 100$



STATE x_{i2} SYNCHRONIZE TO
LEADER'S STATE WHEN $\alpha = 1$
AND $c = 100$



SUMMARY

- Multi-Agent systems
- H_∞ and Adaptive controller
- Synchronization in presence of attacks and disturbances.